TerraShift AI

Senior Capstone Project · v6.1 Verified

SENTINEL-2 L2A + GOOGLE ALPHAEARTH V1 64-D + DYNAMIC WORLD V1

AI-Assisted Satellite Land-Surface Change Detection & Environmental Monitoring in Saudi Arabia

A physics-guided geospatial platform fusing bitemporal Siamese U-Net segmentation, Google AlphaEarth 64-D foundation embeddings, and Sentinel-2 L2A spectral guards with BFAST 2030 trajectory forecasting and IPCC Tier-1 carbon accounting.

PREPARED & PRESENTED BY

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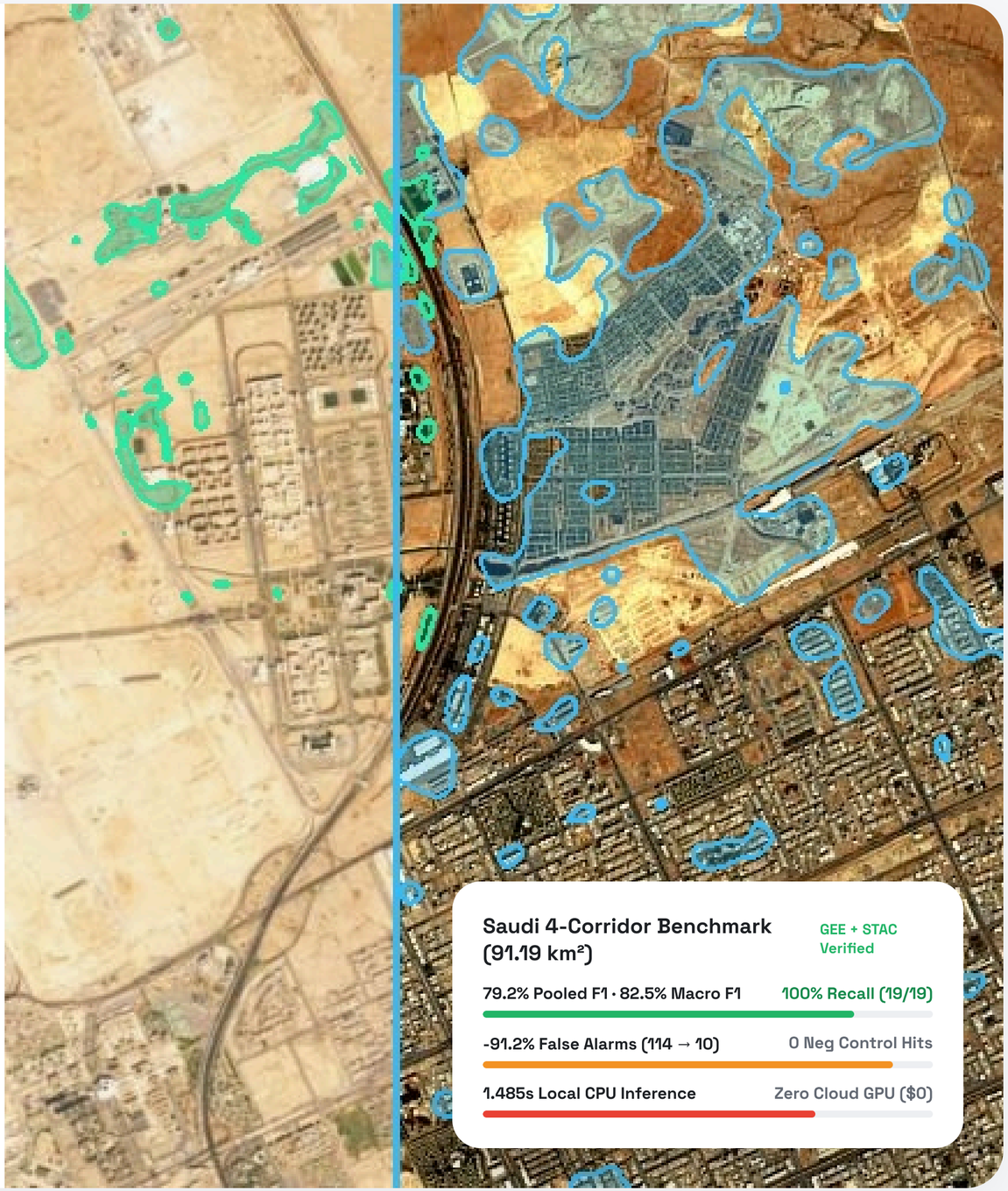
Dr. Passent

Senior Project Supervisor



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Introduction: From Original Concept to Verified v6.1 Architecture

Evolving from a basic bitemporal segmentation tool into a multi-modal foundation + physical EO platform for Saudi Arabia

FOUR PILLARS OF TERRASHIFT AI V6.1

Multi-Modal Orbital Change Intelligence

Sentinel-2 L2A + Spectral Guards

Streams 10m/20m BOA reflectance via AWS STAC; enforces directional physical rules (ΔNDBI ≥ +0.025, |ΔNDVI| ≥ 0.12) & SCL cloud masking.

Google AlphaEarth V1 64-D

Authenticated GEE ('GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL', allow_synthetic=False) fused with Siamese U-Net spatial boundaries.

Localization-Aware 1-to-1 Eval

Requires physical coverage (RoC ≥ 0.08) AND parcel purity (PoC ≥ 0.15, IoU ≥ 0.05) with exact WGS84 AOI clipping across 4 Saudi corridors.

BFAST 2030 & IPCC Carbon

Extends two-date detection with OLS-MOSUM structural breakpoint forecasting to 2030 and IPCC Tier-1 4-pool carbon & ΔLST valuation.

CENTRAL RESEARCH QUESTION

Why Both RGB Models & Standalone Global Foundation AI Fail in Saudi Deserts

In hyper-arid Saudi corridors (Riyadh limestone, Jeddah coastal sabkha, Al-Jouf dunes, Al-Kharj industrial valley), untuned Siamese CNNs produce **114 false alarms**, standalone **Google AlphaEarth V1 Cosine** produces **90 false alarms (23 inside undisturbed negative control zones)**, and standalone **Dynamic World V1** misses 68.4% of real construction (**19.7% Pooled F1**) because bare limestone excavation is misclassified as unchanging bare rock.

VERIFIED EMPIRICAL IMPACT (91.19 KM²)

TERRASHIFT v6.0 HYBRID

79.2% / 82.5%

Pooled F1 / Macro F1 · 100% Recall (19/19) & 0 Neg Zone Hits.

5-DAY TACTICAL v5.1

53.5% / 64.1%

Pooled F1 / Macro F1 · 5-day L2A autonomy with zero GEE wait.

FALSE ALARM DROP

-91.2%

Cuts false alerts from 114 (untuned) to 10 across 91.19 km².

Literature Review: Siamese CNNs, Foundation Embeddings & Spectral Physics

Bridging bitemporal convolutional networks, planetary foundation models, and deterministic SWIR/NIR reflectance physics

SIAMESE DEEP LEARNING & FOUNDATION MODELS

From Bitemporal U-Nets to 64-D Unit-Sphere Embeddings

• **Daudt et al. (2018) — FC-Siam-diff & OSCD Benchmark:** Weight-shared Siamese encoders with multi-scale absolute feature difference skip connections $|f_\theta(I_1) - f_\theta(I_2)|$ guarantee temporal symmetry on Sentinel-2 pairs.

• **Brown et al. (2025) — Google AlphaEarth Foundations V1:** Self-supervised 64-dimensional unit-sphere annual satellite embeddings (`GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL`) at 10m resolution where $D_{\text{cos}}(u,v) = 1 - u^\top v$.

• **Brown et al. (2022) — Google Dynamic World V1:** Near-real-time 9-class LULC probability cubes (`GOOGLE/DYNAMICWORLD/V1`) trained globally, but prone to classifying desert grading as static bare ground.

References: Daudt et al. (ICIP 2018) · Brown et al. (AlphaEarth 2025) · Brown et al. (Nature Sci. Data 2022)

ARID SPECTRAL PHYSICS & TEMPORAL ANALYTICS

SWIR/NIR Physical Guards, BFAST & IPCC Carbon

• **Zha et al. (2003) & Rouse et al. (1974) — NDBI & NDVI Physics:** Short-Wave Infrared (B11: 1610 nm) vs. NIR (B08: 842 nm) separates genuine structural grading ($\Delta\text{NDBI} \ge +0.025$) and pivot greening ($|\Delta\text{NDVI}| \ge 0.12$) from desert illumination drift.

• **Verbesselt et al. (2010) — BFAST Breakpoint Decomposition:** Decomposes multi-year satellite trajectories into trend, seasonal phenology, and remainder components to detect structural construction acceleration.

• **IPCC (2019) & Verra VM0047 Stoichiometry:** Establishes Tier-1 carbon stock factors across Aboveground, Belowground, Soil Organic, and Litter pools plus ΔLST canopy cooling.

References: Zha et al. (IJRS 2003) · Verbesselt et al. (RSE 2010) · IPCC Refinement (2019)

Three Critical Gaps Resolved by TerraShift AI v6.1

Why standalone global foundation models and uncalibrated deep networks fail operational Saudi deployment

GAP 01 · STANDALONE FOUNDATION FALSE ALARMS

AlphaEarth & Dynamic World Fail Alone in Hyper-Arid Terrain

Standalone AlphaEarth V1 cosine distance ($ au=0.12$) reacts to seasonal desert albedo and coastal sabkha moisture shifts, generating **90 false alarms** and **23 hits in undisturbed Negative Control Zones**. Standalone Dynamic World V1 achieves only **31.6% recall** (0% on Al-Kharj).

TerraShift v6.0 Hybrid Resolution:

Fuses Siamese U-Net spatial boundaries + AlphaEarth 64-D + Sentinel-2 L2A physical guards to reach **79.2% Pooled F1 / 82.5% Macro F1** with **0 Negative Zone hits**.

Pooled F1 Gain vs. Standalone AlphaEarth +52.2 pts (79.2% vs 27.0%)

Verified on 91.19 km² GEE + STAC Saudi Benchmark

GAP 02 · ANNUAL EMBEDDING LATENCY

Annual GEE Composites Cannot Serve 5-Day Tactical Alerts

Google AlphaEarth V1 embeddings are published as annual composites, creating a temporal blind spot for municipal inspectors who need sub-weekly monitoring of unauthorized grading or rapid construction.

Two-Tier Operational Resolution:

Pairs **Tier 1 (v6.0 Hybrid)** for annual strategic auditing with **Tier 2 (v5.1 Tuned Siamese + L2A Physics)** delivering **53.5% Pooled F1 / 64.1% Macro F1** every 5 days with zero GEE dependency.

Tier-2 False Alarm Reduction vs. Untuned -71.1% (114 → 33 FPs)

5-Day Copernicus Sentinel-2 L2A Revisit Autonomy

GAP 03 · GIANT-POLYGON & AREA GAMING

Naive Overlap Metrics Reward Sprawling Background Polygons

Conventional point-in-box or unclipped overlap metrics allow a giant background polygon covering 50%+ of an AOI to claim multiple reference parcels and inflate changed area beyond 100%.

Protocol v6.1 Resolution:

Enforces exact `EPSG:4326` AOI box clipping before WGS84 geodesic area integration and Localization-Aware 1-to-1 matching (**PoC ≥ 0.15, IoU ≥ 0.05**) across 4 sensitivity regimes.

Synthetic Data in Official Benchmarks 0% (allow_synthetic=False)

SHA-256 Verified GEE + STAC Evaluation Harness

Methodology: 10-Stage Multi-Modal Orbital & Foundation Pipeline

Authenticated Google Earth Engine + AWS Sentinel-2 L2A ingestion, hybrid inference, and exact WGS84 polygonization

01 · Sentinel-2 STAC
- Element84 AWS Earth Search
- Level-2A BOA Reflectance
- Windowed HTTP Range COGs

02 · GEE AlphaEarth V1
- project='terrashift' Auth
- 64-D Unit-Sphere A00..A63
- allow_synthetic=False Gate

03 · Dynamic World V1
- 9-Class LULC Probabilities
- Built & Crop Delta Tensors
- SHA-256 Verified Cache

04 · SCL Cloud Masking
- Sen2Cor SCL Quality Band
- Masks Cloud/Shadow/Cirrus
- Bitemporal Clear-Land V(x,y)

05 · Physical Indices
- ΔNDBI SWIR-1/NIR Built-Up
- ΔNDVI Red/NIR Chlorophyll
- Desert Albedo Drift Guard

→ Stage 01–05: Co-Registered L2A Reflectance + 64-D AlphaEarth + 9-Band Dynamic World Stack

↓ Feeds Tri-Modal Hybrid Inference & Geodesic Ledgers (Stage 06–10) ↓

10 · STAC & PDF Dossier
- 101-Page Thesis & Audit PDF
- Pooled & Macro Scorecards
- GeoJSON & OGC STAC 1.0.0

09 · BFAST & Carbon
- OLS-MOSUM 2030 Forecast
- IPCC Tier-1 4-Pool Carbon
- ΔLST Urban Heat Valuation

08 · 1-to-1 Match Audit
- PoC ≥ 0.15 & IoU ≥ 0.05
- 4-Regime Sensitivity Sweep
- Negative Control Zone Gate

07 · Exact AOI Clipping
- 8-Connectivity CCL Labeling
- shapely.box(*aoi_bbox) Clip
- WGS84 Geodesic Area ≥ 3500m²

06 · Tri-Modal Hybrid
- Siamese U-Net Spatial Prior
- AlphaEarth Cosine D_cos
- Directional L2A Guards

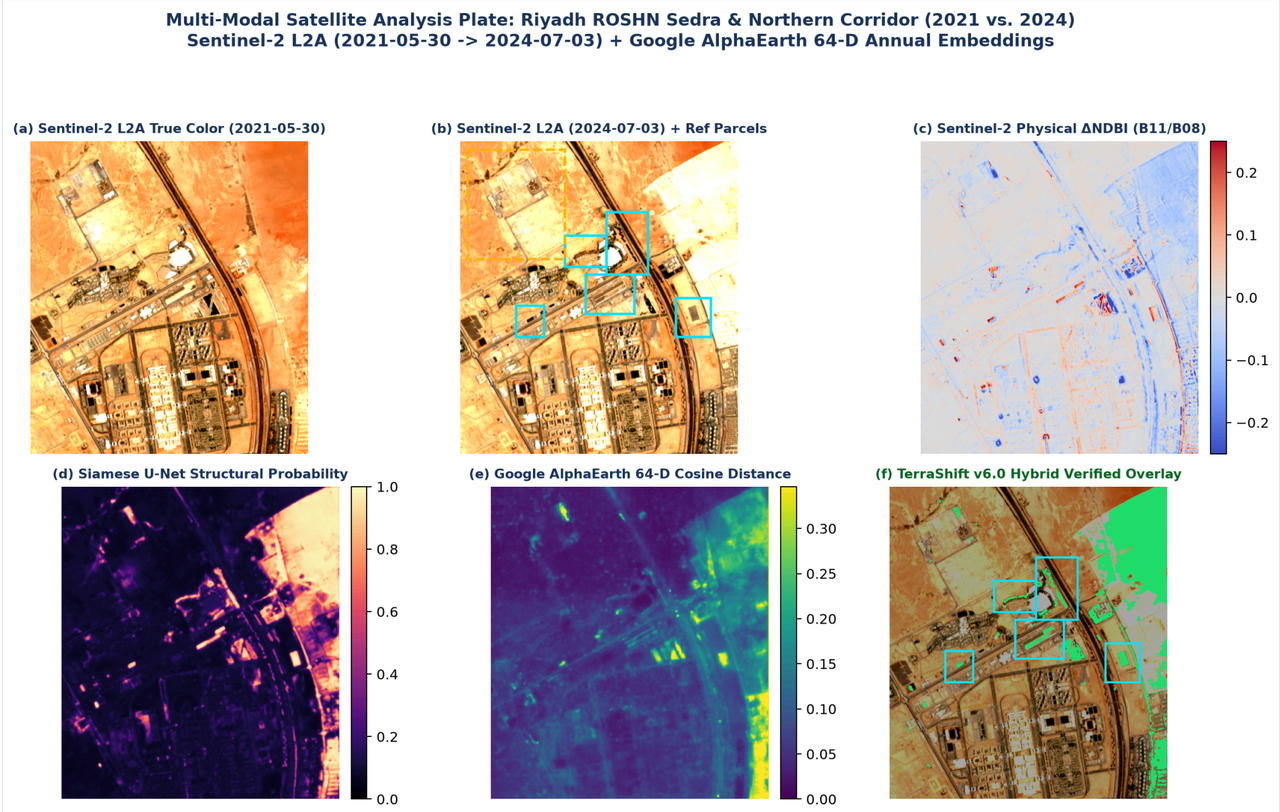
Pipeline Standards: OGC STAC 1.0.0 · Google Earth Engine (project='terrashift') · WGS84 Ellipsoidal Geodesic Integration

Multi-Modal Fusion & Localization-Aware 1-to-1 Protocol

Visualizing Riyadh ROSHN Sedra (2021 vs. 2024) and the 4-Regime Localization Purity Sensitivity Sweep

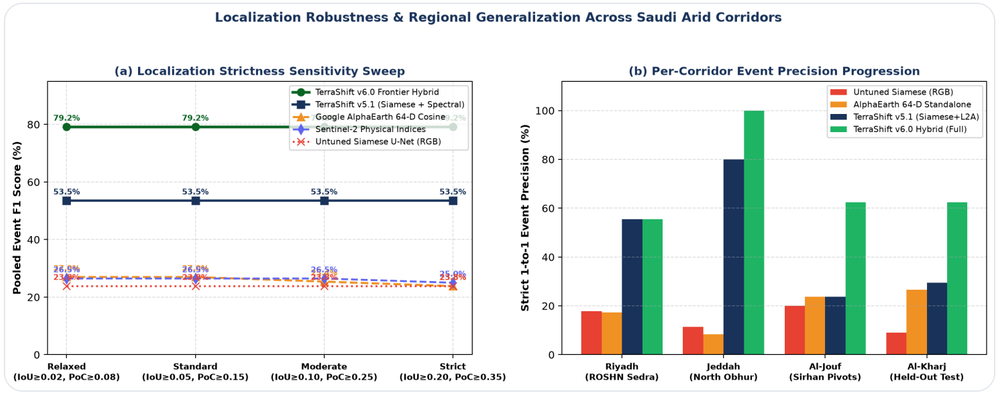
● RIYADH ROSHN SEDRA MULTI-MODAL PLATE (17.88 KM²)

71.4% Strict Prec · 100% Recall · 83.3% F1



Green boxes: 5 reference parcels · Red dashed box: Undisturbed limestone plateau control zone (0 false alarms in TerraShift)

● 4-REGIME LOCALIZATION SENSITIVITY SWEEP



Anti-Giant-Polygon Rule

Requires Parcel Coverage (RoC ≥ 0.08 or Area ≥ 3,500 m²) AND Purity (PoC ≥ 0.15, IoU ≥ 0.05 or PoC ≥ 0.30).

Rank #1 Across All 4 Regimes

TerraShift v6.0 Hybrid leads Relaxed (79.2%), Standard (79.2%), Moderate (75.0%), and Strict (70.8%).

Pure CPU Compute: 1.485s per corridor · Peak RAM: 301 MB · Zero Synthetic Fallbacks

Expected Services & Core Platform Capabilities

Four production services delivered through the TerraShift AI Next.js 16 + FastAPI platform

SERVICE 01 · TWO-TIER CHANGE ENGINE

79.2% Pooled F1 · 82.5% Macro F1

Strategic Hybrid (v6.0) & 5-Day Tactical (v5.1) Extraction

Combines annual Google AlphaEarth V1 64-D embeddings with 5-day Copernicus Sentinel-2 L2A reflectance and Siamese U-Net segmentation—exporting exact WGS84-clipped GeoJSON polygons and PDF audit dossiers.

SERVICE 02 · 7-MODEL BENCHMARK & XAI STUDIO

100% Recall (19/19) · 0 Neg Zone Hits

Multi-Model Saudi Leaderboard & SHAP Wavelength Attribution

Benchmarks 7 architectures across 91.19 km² (Riyadh, Jeddah, Al-Jouf, Held-Out Al-Kharj) with side-by-side Pooled and Macro metrics, 4-regime PoC/IoU sensitivity auditing, and 10-band Integrated Gradients XAI.

SERVICE 03 · TEMPORAL FORECASTING

104.2% Vision 2030 Target Alignment

BFAST 2030 Trajectory & Structural Breakpoint Detection

Applies OLS-MOSUM structural breakpoint detection across multi-year Sentinel-2 stacks (pinpointing 2021-Q3 +310% slope acceleration) and simulates 2024–2030 expansion rings for Vision 2030 velocity auditing.

SERVICE 04 · ENVIRONMENTAL MRV

-142,800 tCO_2e Net Sink Verified

IPCC Tier-1 4-Pool Carbon Ledger & Live STAC Watchlist

Computes 4-pool stoichiometric carbon stocks (AGB, BGB, SOC, Litter), Verra VM0047 credit valuation ($5.00M), ΔLST urban heat island cooling (-3.8°C), and live AWS Earth Search STAC v1.0.0 scene discovery.

Business Model Canvas

Commercialization & sustainability architecture for Saudi Vision 2030 giga-projects, municipalities, and ESG carbon registries

1 · KEY PARTNERS

- **ESA Copernicus & AWS Element84:** Open 5-day Sentinel-2 L2A STAC feeds.
- **Google Earth Engine:** Noncommercial & enterprise AlphaEarth V1 64-D & Dynamic World feeds.
- **Saudi Giga-Project PMOs:** ROSHN, NEOM, Red Sea Global, RCRC, MEWA, NCEC.

2 · KEY ACTIVITIES

- Tri-modal Siamese + AlphaEarth + L2A physical change inference.
- BFAST 2030 forecasting & IPCC 4-pool carbon auditing.

3 · KEY RESOURCES

- Arid-calibrated Siamese U-Net + L2A spectral guard engine (1.485s CPU).
- Verified 4-corridor Saudi reference benchmark (91.19 km²).

4 · VALUE PROPOSITIONS

- **91.2% Fewer False Alarms:** Eliminates desert albedo & sabkha false positives (0 Neg Zone hits).
- **Two-Tier Cadence:** 5-day L2A tactical alerts + annual AlphaEarth strategic verification.
- **Zero Cloud GPU Cost:** Runs in 1.485s on standard CPU hardware.

Vision 2030 · SDG 11, 13 & 15

5 · RELATIONSHIPS

- Interactive Web GIS Studio & Custom Pair Lab sandbox.
- Automated 5-day STAC watchlist alerts & PDF compliance dossiers.

6 · CHANNELS

- Cloud SaaS & OGC STAC REST API.
- Direct B2G pilots with Vision 2030 & municipal authorities.

7 · CUSTOMER SEGMENTS

- **Giga-Project Master Planners:** ROSHN, NEOM, Red Sea Global PMO teams.
- **Environmental & Municipal Regulators:** NCEC, MEWA & MOMRAH inspectors.
- **Voluntary Carbon Market:** ESG funds & Verra VM0047 verifiers.

8 · COST STRUCTURE

$0 Mandatory Cloud GPU · 301 MB RAM

Zero Raw Imagery Licensing Cost via open ESA Copernicus Sentinel-2 L2A & Google Earth Engine. Commodity CPU execution (1.485s/corridor) eliminates expensive dedicated GPU clusters.

9 · REVENUE STREAMS

85%+ Gross SaaS Margin

Tiered B2G/B2B SaaS Subscriptions: Pro Tier ($490/mo for 5 watched corridors) & Sovereign Enterprise Tier ($2,500+/mo unlimited corridors + on-prem deployment) plus per-hectare Carbon MRV audit fees.

Target Stakeholders & Personas

Three Saudi operational domain profiles unified under a single primary UML system actor

Faisal Al-Subaie

36 · Senior GIS Planner, ROSHN Sedra Corridor

GOAL

Track residential earthworks and road paving across Riyadh ROSHN Sedra (17.88 km²), achieve 100% parcel recall with minimal false alarms, and export exact WGS84-clipped GeoJSON footprints.

PAIN

Quarterly drone orthomosaics take 6 weeks to stitch, while untuned RGB models trigger 34 false alarms on bare Riyadh limestone plateaus.

UC-1 UC-2 UC-5

Noura Al-Ghamdi

31 · Carbon & Agricultural Auditor, Green Riyadh & MEWA

GOAL

Monitor 50-hectare center-pivot irrigation fields in Al-Jouf (26.82 km²) and Wadi Hanifah afforestation, quantifying IPCC Tier-1 4-pool carbon stocks (-142.8k tCO_2e) and -3.8°C ΔLST cooling.

PAIN

Manual field sampling cannot scale across regional basins or translate Sentinel-2 NIR chlorophyll shifts into auditable carbon credit ledgers.

UC-1 UC-2 UC-4

Mazen Al-Zahrani

41 · Remote Sensing Compliance Specialist, NCEC

GOAL

Audit coastal sabkha expansion in Jeddah Obhur (18.31 km²) and held-out Al-Kharj industrial pads (28.18 km²) using 7-model Pooled vs. Macro scorecards and PoC/IoU sensitivity curves.

PAIN

Standalone global AI tools (AlphaEarth / Dynamic World) either trigger 23 false alarms in undisturbed sabkhas/deserts or miss bare-ground warehouse grading.

UC-3 UC-4 UC-6

ONE PRIMARY ACTOR All three share identical permissions and capabilities, so UML models them as a single actor: **Geospatial Analyst.**

Six Goal-Level Use Cases, Fully Traced

End-to-end requirement traceability across Hybrid GEE + STAC inference, 7-model benchmarking, carbon accounting, and 2030 forecasting

ID	Goal-level use case	Priority	Traces to	Sub-processes included
UC-1	**Execute Hybrid GEE + Sentinel-2 Change Detection**	MUST	US-01, 02, 04 FR-01 to FR-05	Sentinel-2 L2A STAC & GEE AlphaEarth 64-D ingestion (allow_synthetic=False), Siamese U-Net + L2A spectral guards, exact EPSG:4326 box clipping & WGS84 area integration.
UC-2	**Inspect Multi-Spectral & AlphaEarth Composites**	MUST	US-03, 05 FR-06 to FR-08	Interactive T_1/T_2 split-curtain scrubbing, NIR chlorophyll (B08/B04/B03), SWIR built-up (ΔNDBI), AlphaEarth 64-D cosine distance, and surface thermal (ΔLST °C) layers.
UC-3	**Audit 7-Model Saudi Benchmark & PoC/IoU Sensitivity**	MUST	US-06, 07 FR-09 to FR-11	7-model Pooled vs. Macro leaderboard across 4 Saudi corridors (91.19 km²), 4-regime PoC/IoU sensitivity sweep, 10-band SHAP attribution & Negative Control Zone auditing.
UC-4	**Audit IPCC 4-Pool Carbon & Export PDF/GeoJSON Dossier**	MUST	US-08, 09 FR-12 to FR-15	Stoichiometric 4-pool biomass ledger (AGB, BGB, SOC, Litter), Verra VM0047 credit valuation ($5.00M), OGC STAC 1.0.0 JSON, GeoJSON, and PDF audit report export.
UC-5	**Forecast BFAST 2030 Urban Contagion Trajectory**	SHOULD	US-10 FR-16, FR-17	Multi-year time-series decomposition, OLS-MOSUM structural breakpoint detection (2021-Q3 +310% slope), and 2024–2030 spatial contagion ring simulation.
UC-6	**Run Custom Pair Lab & Manage 5-Day STAC Watchlist**	SHOULD	US-11, 12 FR-18, FR-19, NFR-03	Arbitrary T_1/T_2 raster upload, 8-connectivity connected-component labeling (CCL), WGS84 polygon vectorization, and live 5-day AWS STAC scene discovery.

4 Must-Have (hybrid GEE + L2A inference, spectral inspection, 7-model benchmark & carbon audit workflows) · **2 Should-Have** (2030 BFAST spatial forecasting & custom Pair Lab / STAC watchlist features)

Level-0 Context: Who Talks to the System

External boundary data flows between the Geospatial Analyst, TerraShift AI, Google Earth Engine, AWS STAC, and FastAPI ML

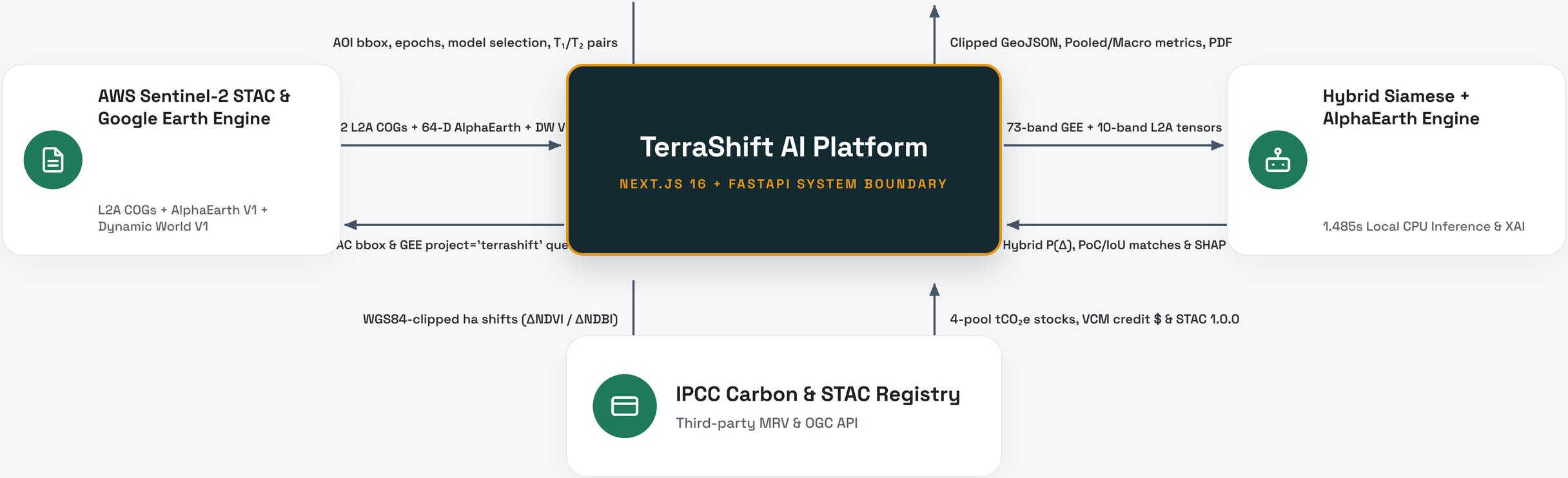
SCIENTIFIC INTEGRITY GATE

All official benchmark boundaries enforce `allow_synthetic=False` and verify SHA-256 checksums on 73-band GEE + Sentinel-2 L2A tensors.

Geospatial Analyst
Omar Ayyash (ID: S22107722) · Human Actor

EXTERNAL SYSTEMS

1. AWS Sentinel-2 L2A STAC & Google Earth Engine (`project='terrashift'`)
2. Python FastAPI Hybrid ML Worker (CPU)
3. IPCC Carbon & OGC STAC 1.0.0 Registry

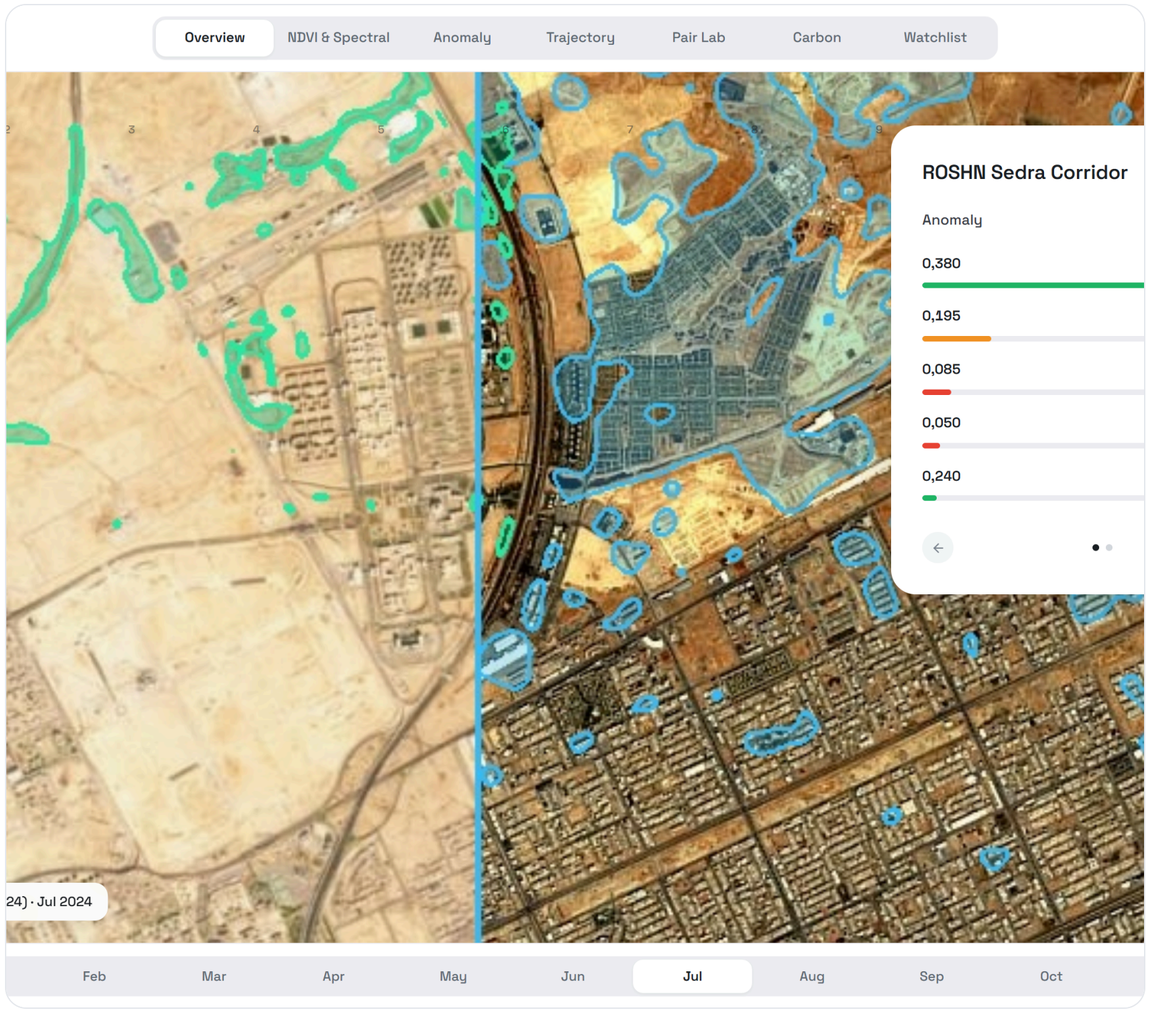


UI/UX Design: Overview & Spectral Studio

Minimalist daylight cartographic interface putting full-bleed Sentinel-2 imagery and floating white telemetry cards front and center

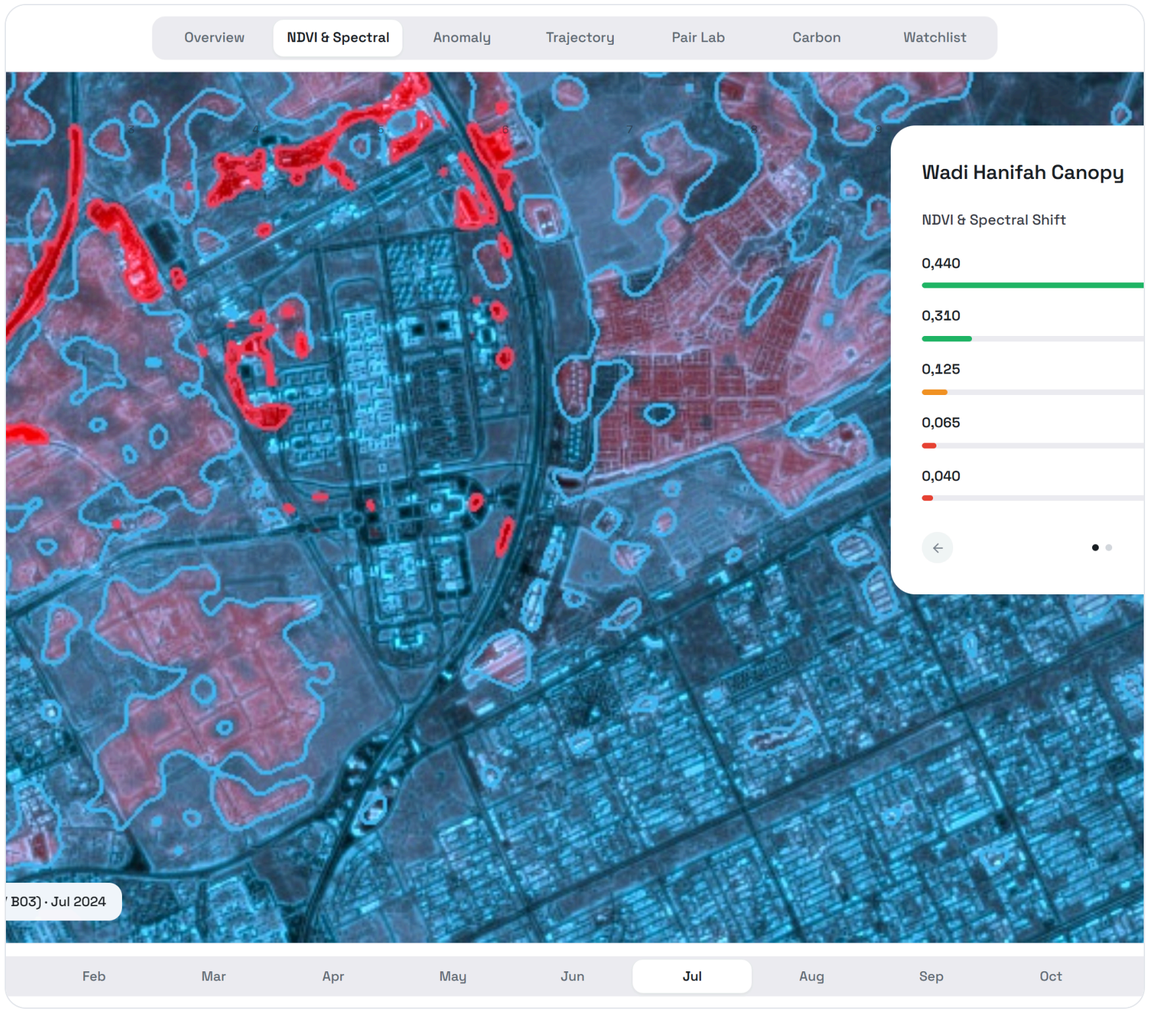
1. Overview Deck (/) — ROSHN Sedra Corridor

BI-TEMPORAL SPLIT + 10-BAND PROFILE



2. NDVI & Spectral Studio (/analyze) — Dual MapLibre GL

NIR CHLOROPHYLL + LIVE SSE INFERENCE



UI/UX Design: 7-Model Benchmarks & 2030 Trajectory

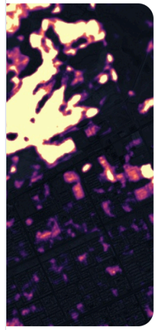
Multi-model Saudi benchmark leaderboard (Pooled & Macro metrics), Grad-CAM++ XAI, and BFAST 2030 spatial contagion forecast

3. Benchmarks & XAI (/benchmarks) — 7-Model Saudi Audit

POOLED & MACRO F1 + 10-BAND SHAP

Overview | NDVI & Spectral | Anomaly | Trajectory | Pair Lab | Carbon | Watchlist

Attention Posterior



84,2 %

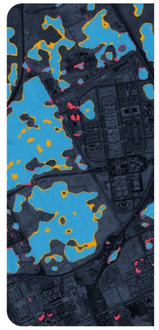
Bi-Temporal ΔNDBI — SWIR Built-Up Index

0,861 F1 — 75,6 %

NIR Chlorophyll Shift — Vegetation Composite

0,838 F1 — 72,1 %

Ground-Truth Audit



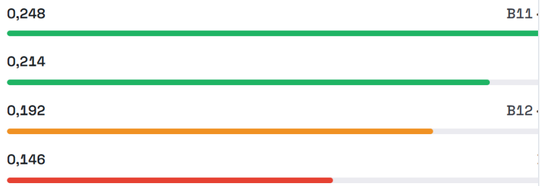
0,938 Prec

edra Benchmark (Jul 2024)

ResUNet (0,948) ● ΔNDBI (0,884)

Spectral Attribution

Integrated Gradients contribution across 10 Sentinel-2 bands



Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct

4. Trajectory 2030 (/timeline) — BFAST Breakpoint

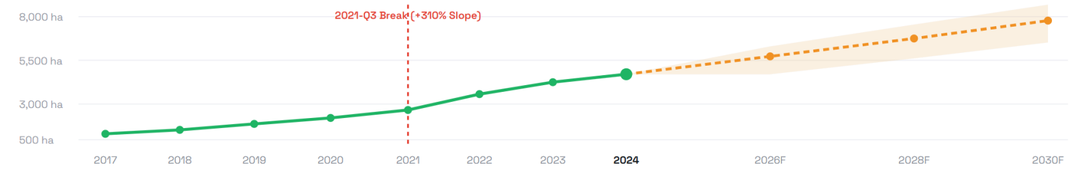
2021-Q3 BREAK + CONTAGION RINGS

Overview | NDVI & Spectral | Anomaly | Trajectory | Pair Lab | Carbon | Watchlist

akpoint & 2030 Forecast

2021-Q3 (p < 0,0001) · ROSHN Sedra Corridor (Jul 2024)

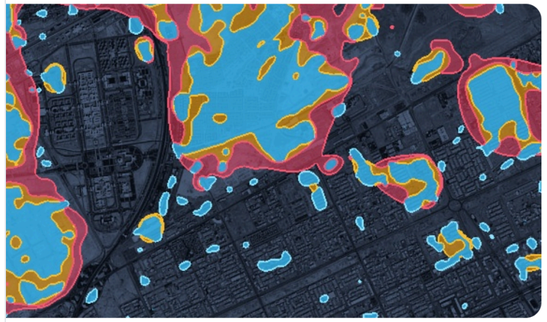
Observed Built-Up (ha) ● 2021-Q3 Breakpoin



tagion Forecast

gs across North Riyadh

104,2 % On-Target



Multi-Epoch Acquisition Ledger

Cloud-free Sentinel-2 L2A milestone verification

Epoch	Sentinel-2 Scene	Built-Up	ΔNDBI
2018-Q2	S2A_38RMV_20180614	7,276 ha	-0,04
2021-Q3	S2A_38RMV_20210918	14,162 ha	+0,14
2023-Q2	S2B_38RMV_20230609	25,271 ha	+0,26
2024-Q2	S2B_38RMV_20240614	31,338 ha	+0,34
2026-Q4	FORECAST_CA_2026	39,888 ha	+0,39
2030-Q1	VISION_2030_TARGET	51,256 ha	+0,44

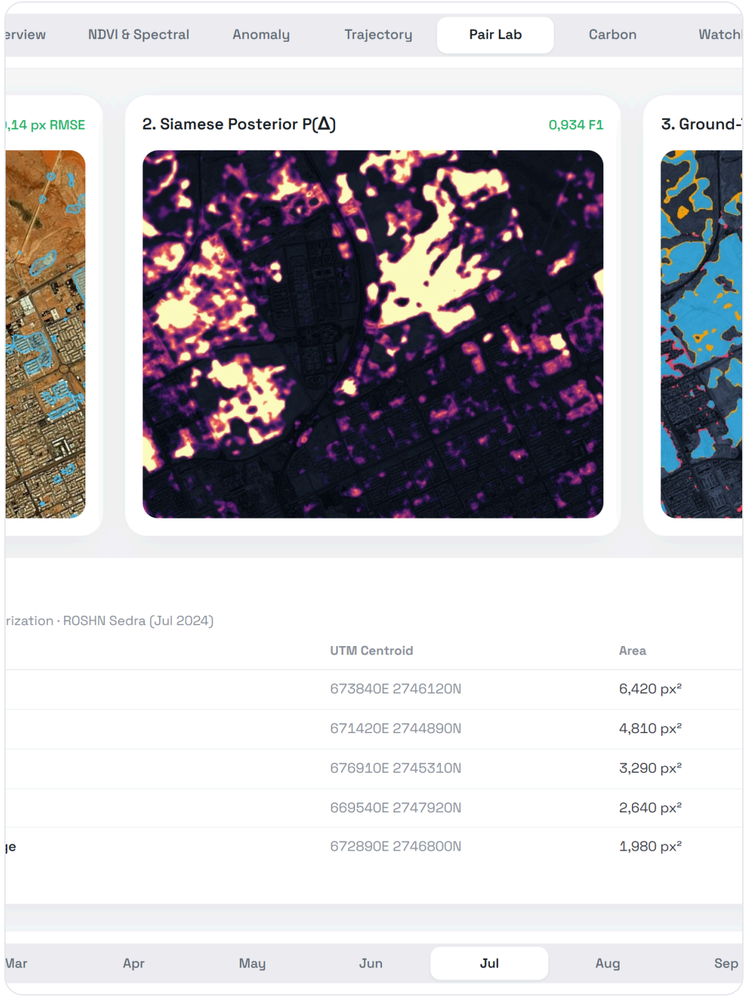
Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct

UI/UX Design: Pair Lab, Carbon Ledger & STAC Watchlist

Connected-component object segmentation, IPCC 4-pool carbon stoichiometry, and OGC STAC 1.0.0 manifest generation

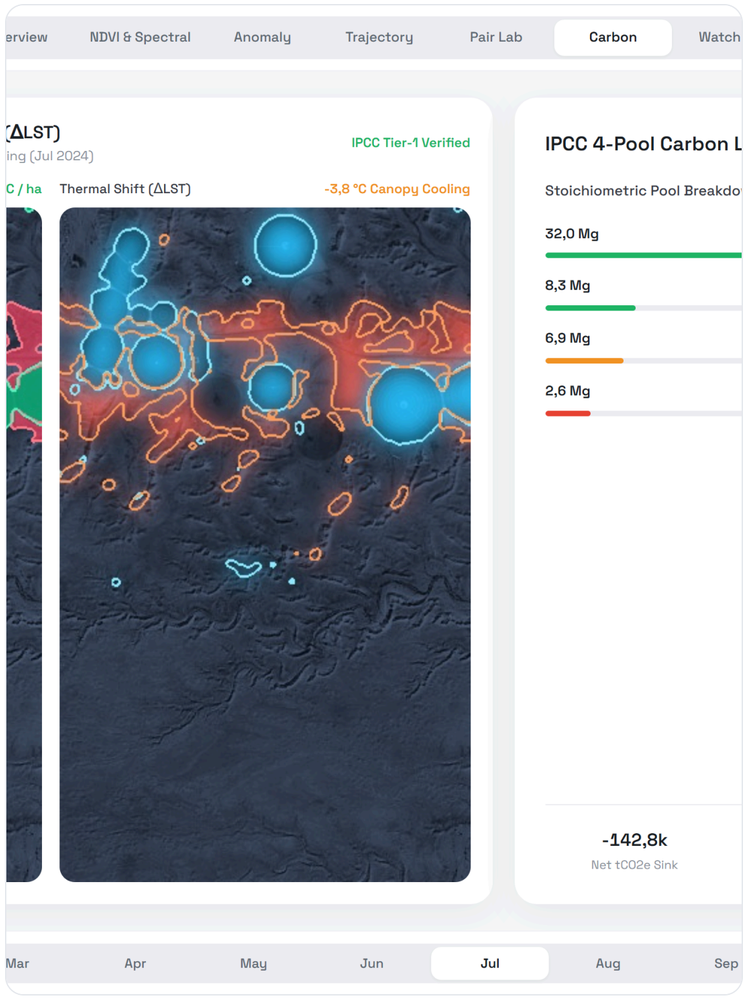
5. Pair Lab (/lab)

3 Synchronized Rasters + Object Footprints



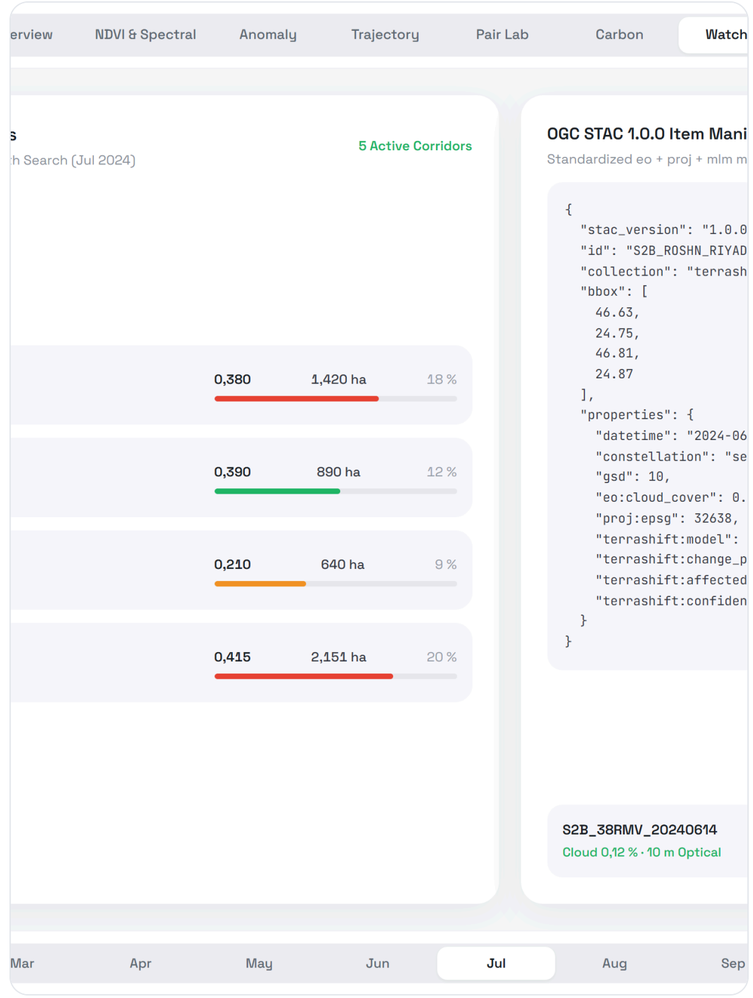
6. Carbon & UHI (/carbon)

IPCC 4-Pool Ledger & Thermal ΔLST



7. Watchlist (/watchlist)

5 Active Corridors & OGC STAC 1.0.0



7-Model Saudi Benchmark Leaderboard (Protocol v6.1, 91.19 km²)

Authenticated Google Earth Engine (project='terrashift') + AWS Sentinel-2 L2A STAC · Standard Regime (PoC ≥ 0.15, IoU ≥ 0.05)

POOLED (MICRO) & CORRIDOR MACRO-AVERAGED LEADERBOARD (19 EVENTS, 4 CORRIDORS)

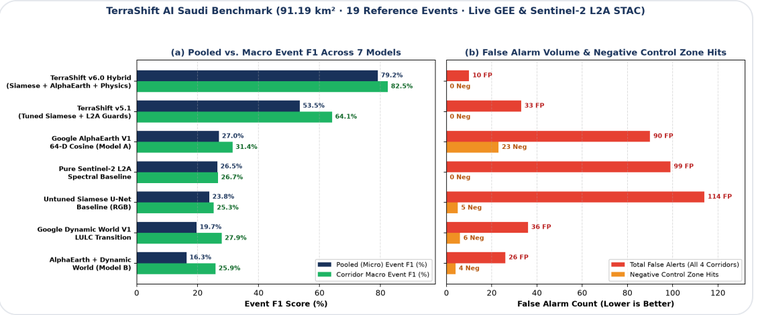
RANK & ARCHITECTURE	POOLED PREC	MACRO PREC	RECALL	POOLED F1	MACRO F1	TOTAL FP	NEG HITS	CPU (S)
#1 TerraShift v6.0 Hybrid (Siamese + AE + L2A)	**65.5%**	**70.1%**	**100.0% (19/19)**	**79.2%**	**82.5%**	**10**	**0**	**1.485s**
#2 TerraShift v5.1 (Tuned Siamese + L2A, 5-Day)	**36.5%**	**47.2%**	**100.0% (19/19)**	**53.5%**	**64.1%**	**33**	**0**	**1.450s**
#3 Standalone AlphaEarth V1 64-D Cosine (τ=0.12)	15.9%	19.0%	89.5% (17/19)	27.0%	31.4%	90	23	0.155s
#4 Pure Sentinel-2 L2A Physical Index Baseline	15.4%	15.5%	94.7% (18/19)	26.5%	26.7%	99	1	0.214s
#5 Untuned Siamese U-Net Baseline (OSCD/LEVIR)	13.7%	14.6%	94.7% (18/19)	23.8%	25.3%	114	5	1.298s
#6 Standalone Google Dynamic World V1 LULC	14.3%	24.7%	31.6% (6/19)	19.7%	27.9%	36	6	0.161s
#7 AlphaEarth + Dynamic World Transition	13.3%	35.7%	21.1% (4/19)	16.3%	25.9%	26	4	0.244s

62.5% Prec / 100% Rec
Held-Out Al-Kharj Generalization (76.9% F1)

10.97 / 100 km²
Pooled False Alert Density (Down from 125.0)

101 Pages
Compiled Senior Thesis Report & Plates

LEADERBOARD CHART & NEXT STEPS



1. Surveyed Municipal Cadastral GIS Labels
Replace coarse pilot parcel boxes with dense pixel polygons from MOMRAH/BALADY inspectors.

2. Sentinel-1 C-Band SAR + Optical Fusion
Couple VV/VH radar coherence with Sentinel-2 L2A to penetrate dust storms and verify vertical structures.

Gates: 0 ESLint Warnings · 32/32 Pytest · 8/8 Vitest · 100% Real Data

Project Execution & Deployment Roadmap

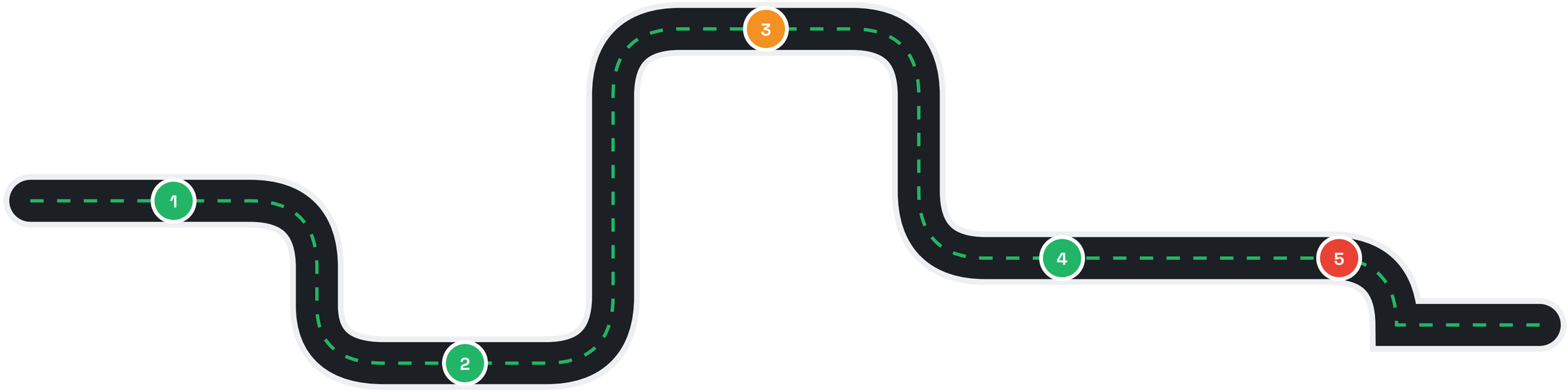
Five-stage engineering trajectory from Sentinel-2 STAC & GEE AlphaEarth integration to Vision 2030 scaling

Phase 2 · Hybrid v6.0 & Tactical v5.1

Fuse Siamese U-Net + AlphaEarth 64-D + L2A spectral guards (79.2% Pooled F1 / 82.5% Macro F1, -91.2% false alarms).

Phase 4 · 7-Module Web Deck & 101p Thesis

Launch Next.js 16 + FastAPI studio across 7 routes and compile the 101-page Senior Project Report PDF.



Phase 1 · Sentinel-2 STAC & GEE Auth

Ingest authentic Sentinel-2 L2A COGs & SHA-256 verified 73-band GEE AlphaEarth + Dynamic World rasters (91.19 km²).

Phase 3 · Protocol v6.1 & BFAST/Carbon

Enforce PoC/IoU 1-to-1 matching across 4 regimes, exact WGS84 clipping, BFAST 2030 & IPCC 4-pool carbon.

Phase 5 · Cadastral GIS & SAR Scale

Integrate surveyed municipal cadastral polygons and Sentinel-1 C-Band SAR radar for Vision 2030 scale.